

WASIF KARIM

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SUMMARY

Backend-focused software engineer and startup co-founder/CTO who builds production services and AI features: Java/Kotlin microservices at Airbnb, 6 Python/PostgreSQL REST APIs at Hilton, and on-device AI at CWRU. Depth in REST API design, relational data modeling, and full-stack TypeScript with test coverage. B.S. Astrophysics & B.A. Computer Science, GPA 3.82, Honors.

EDUCATION

Ohio Wesleyan University — Delaware, OH

May 2026

B.S. Astrophysics & B.A. Computer Science · GPA 3.82 · Honors

Relevant coursework: Data Structures & Algorithms, Databases, Software Engineering, Machine Learning, Computer Vision, NLP, Numerical Methods.

EXPERIENCE

Robotics Software Engineer — Case Western Reserve University

Jun 2026 – Present

- Porting Poetry Camera (an open-source, Claude-powered AI camera) from a cloud LLM API to a model running locally on a Raspberry Pi-class board, removing per-call API cost and network dependence.
- Building the control software for a volumetric resin-printing prototype (with UC Berkeley) that cures a full part in one rotation, synchronizing the projector with a rotating resin stage instead of printing layer by layer.

Co-Founder & CTO — Bytewright (Software Studio)

Mar 2026 – Present

- Co-founded a 5-person software studio; as CTO, lead backend development for client web apps and build AI-powered admin-support and web-management tools.
- Delivered 5 client projects (BringingGenerations, Ohio Wesleyan University, NNCorporations), owning the backend and AI features.

Software Engineering Intern — Airbnb

May 2025 – Aug 2025

- Built backend logic for an opt-in AI host-messaging feature that drafts guest replies from a host's prior answers, on an in-house, locally hosted LLM.
- Shipped 10 production PRs behind feature flags on a 6-engineer team, in Airbnb's Java/Kotlin microservice codebase.

Software Engineering Intern — Hilton

May 2024 – Aug 2024

- Built 6 Python REST API endpoints backed by PostgreSQL for a guest-facing app (mobile check-in, in-stay services) and a Tableau metrics dashboard, shipped through a Git-based CI/CD pipeline.

Software Development Intern (Part-time) — Ohio Wesleyan University

Aug 2024 – May 2026

- Built Python/SQL data-analysis and reporting tooling used by 30+ users across 6 campus departments; rebuilt the alumni website on the BigTree CMS with Python-automated content workflows.

PROJECTS

VersityRooms — Full-Stack Room-Booking Web App

2025

Next.js 14 · TypeScript · Prisma · PostgreSQL · Tailwind / shadcn-ui · Vitest

- Built a type-safe room-booking app with role-based access (student, faculty, admin), six Prisma/PostgreSQL data models, composite indexing, and audit logging.
- Implemented a component UI (Tailwind, shadcn/ui) with unit tests (Vitest) and CI-ready Vercel/Railway deployment.

Autonomous Vehicle — ROS2 Self-Driving Stack

2025

ROS2 Humble · C/C++ · Python · Jetson Orin Nano · Nav2 · Cartographer SLAM · YOLOv8 · ArduPilot

- Engineered the self-driving stack: perception (LiDAR + YOLOv8), SLAM/navigation (Cartographer + Nav2), EKF sensor fusion, and a MAVLink control bridge on a Jetson Orin Nano.

LEADERSHIP & AWARDS

- HackPrinceton — 2nd Place: led a 5-person team to build OnlySwap, a verified student marketplace (React Native, Node.js, Firebase) with 245+ users.
- Founding President, OWU Robotics Club: grew the club to 267 members, raised \$68K in funding, secured a faculty sponsor, and ran hands-on build events and competitions.

SKILLS

Languages: Java, Python, JavaScript/TypeScript, SQL, C/C++

Backend & APIs: REST APIs, Microservices, Node.js, Flask, FastAPI, Prisma, PostgreSQL, data modeling

CS Foundations: Data Structures & Algorithms, Object-Oriented Design, Unit/Integration Testing, Agile/Scrum

Cloud & DevOps: AWS, Docker, Git, CI/CD

Frontend & AI: React, Next.js, TypeScript, LLM & agent tooling, on-device inference (Raspberry Pi), TensorFlow, PyTorch, ROS2